FACULTY OF COMPUTING

SEMESTER 2

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SECP3213-02 BUSINESS INTELLIGENCE

TOPIC: DATA WORKFLOW

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1.0 Workflow

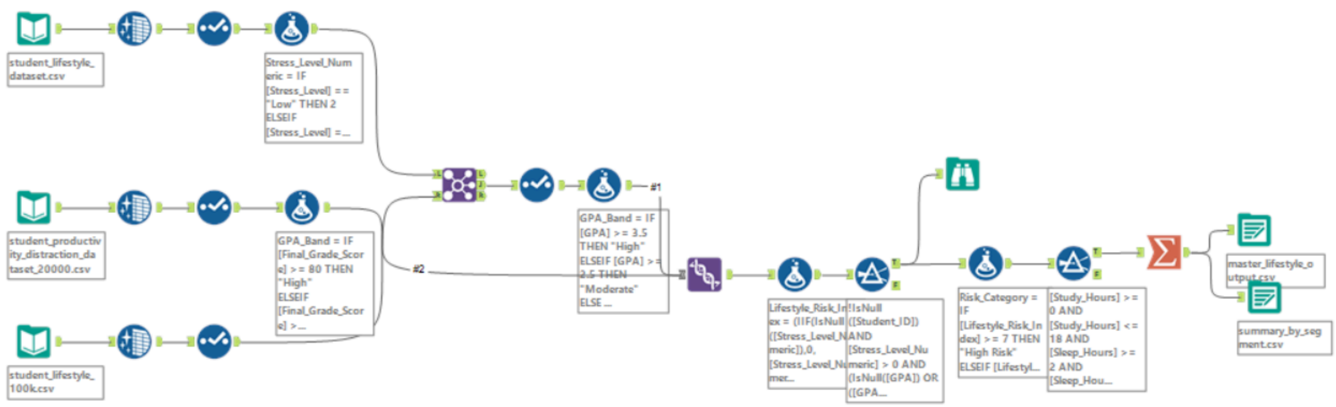


Figure 1: Full Overview of Alteryx Workflow

2.0 Tools Used

1. Data Cleansing Tool

   The Data Cleansing tool was applied as the second step in each of the three tracks, immediately after the Input Data tool. Identical configuration settings were used across all tracks to enforce a uniform data quality baseline:

- Fields Selected: All Fields, ensure no column is unintentionally left with nulls or whitespace regardless of schema variations between datasets.
- Replace Nulls with Blank (String Fields): Enabled, prevents null propagation in categorical columns such as Gender, Department, and Stress_Level.
- Replace Nulls with 0 (Numeric Fields): Enabled, ensures continuity in all quantitative columns (GPA, Study_Hours, Sleep_Hours, etc.) without requiring manual downstream imputation.
- Remove Leading and Trailing Whitespace: Enabled, eliminates hidden space characters that would otherwise break exact match conditions in Formula expressions and downstream join keys.

2. Select Tool

   Following Data Cleansing, a Select tool was connected to each track to enforce the master schema. The Select tool was used to: (1) rename all columns to a consistent PascalCase convention, (2) explicitly cast each column to its correct data type and (3) drop columns not required by the downstream visualization blueprint in order to minimize memory overhead particularly for Track 3. Table 1 shows the detailed changes made to the column names and data types across each data track.

Table 1: New column name and data type

Dataset	Original Column	Renamed To	Data Type
1	Study_Hours_Per_Day	Study_Hours	Double
1	Sleep_Hours_Per_Day	Sleep_Hours	Double

1	Physical_Activity_Hours_Per_Day	Physical_Activity_Hours	Double
2	student_id	Student_ID	Int64
2	study_hours_per_day	Study_Hours	Double
2	stress_level	Stress_Level_Numeric	Int32
2	final_grade	Final_Grade_Score	Double
2	phone_usage_hours	Phone_Usage_Hours	Double
3	CGPA	GPA	Double
3	Sleep_Duration	Sleep_Hours	Double
3	Physical_Activity	Exercise_Minutes	Double
3	Stress_Level	Stress_Level_Numeric	Int32

3. Formula Tool
   (a) Dataset 1 encodes stress levels as text strings. A Formula tool was added immediately after the Select tool in Dataset 1 to create a new derived column, Stress_Level_Numeric (Int32), using the conditional expression in Figure 2.

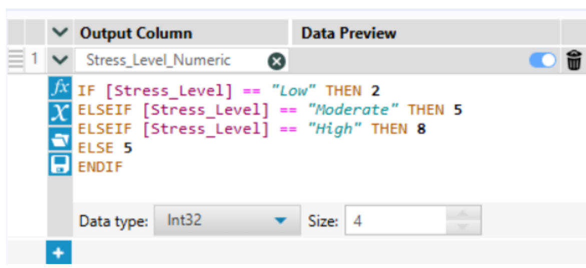


Figure 2: Conditional Expression

This mapping positions Low, Moderate and High at values 2, 5 and 8 respectively on a 1–10 integer scale to directly match the numeric range used by Datasets 2 and 3. The ELSE 5 clause assigns a moderate default for any unexpected or missing string value to prevent null injection into the derived column.

   (b) Used to create the new Risk_Category column by converting the numeric Lifestyle_Risk_Index into readable labels (High Risk, Moderate Risk, Low Risk). It works like an IF-ELSE statement as shown in Figure 3.

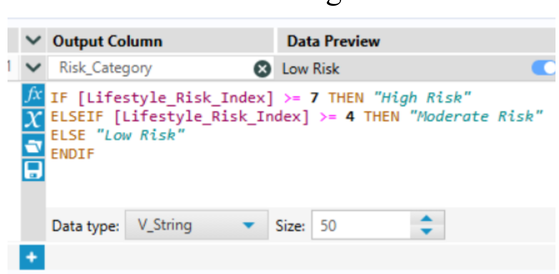


Figure 3: IF-ELSE statement

(c) Two derived features were created. First, Lifestyle_Risk_Index assigns weighted scores across four lifestyle indicators which are Stress_Level_Numeric (40%), insufficient sleep (30%), physical inactivity (20%) and high social hours (10%). Second, Digital_Distraction_Level categorises students as High (6+ hours), Moderate (3–5 hours) or Low (below 3 hours) based on total screen time across Social_Media_Hours, Gaming_Hours and Phone_Usage_Hours.

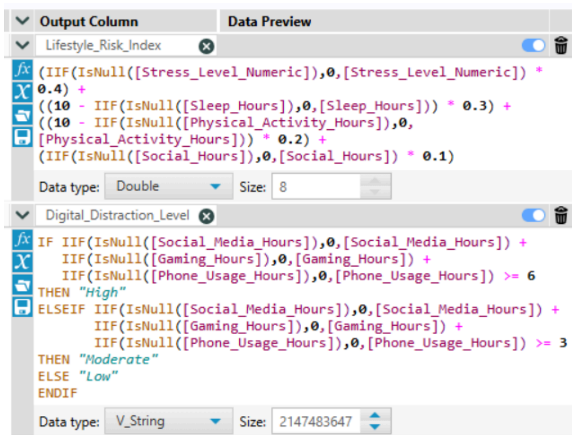


Figure 4 : Feature Engineering Formula Expression

4. Filter Tool

   The tool was applied to remove statistically invalid records from the dataset prior to aggregation and export. A custom expression was configured to retain only records where Study_Hours fell within 0-18, Sleep_Hours within 2-14 and Lifestyle_Risk_Index within 0-1. Records satisfying all conditions were routed through output while invalid records were discarded through the False output.

5. Join Tool

   The Join tool was used to merge Dataset 1 and Dataset 3 using Student_ID as the join key. An inner join was configured so only students present in both datasets were retained. A Select tool was then applied after the join to remove duplicate columns, keeping Track 1 versions as primary while preserving new columns from Dataset 3 such as Depression_Indicator and Social_Media_Hours.

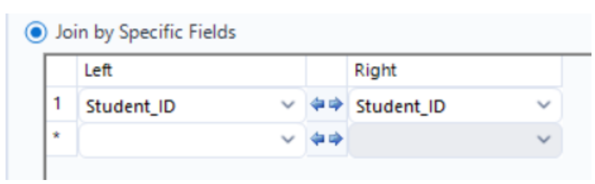


Figure 5 : Join Tool Configuration

6. Union Tool

   Since Dataset 2 uses Final_Grade_Score (0–100) instead of GPA (0–4.0), two Formula tools were configured to create a standardised GPA_Band field on both streams before merging. The Union tool was then set to Auto Config by Name mode to vertically stack both streams into a single master dataset. Columns absent in one stream were populated with NULL values as expected.

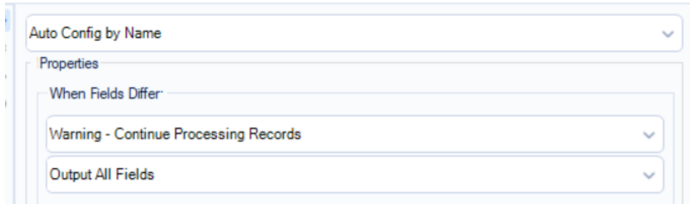
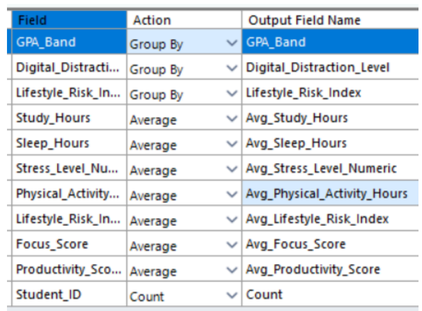


Figure 6 : Union Tool Configuration

7. Summarize Tool

   Used to collapse the master dataset into grouped aggregate rows for segment-level analysis. Figure 4 shows the configuration where GPA_Band, Digital_Distraction_Level and Lifestyle_Risk_Index were set as Group By dimensions, average calculations were applied across seven key lifestyle fields and a Count of Student_ID was included to reflect the population size of each segment.



Field	Action	Output Field Name
GPA_Band	Group By	GPA_Band
Digital_Distracti...	Group By	Digital_Distraction_Level
Lifestyle_Risk_In...	Group By	Lifestyle_Risk_Index
Study_Hours	Average	Avg_Study_Hours
Sleep_Hours	Average	Avg_Sleep_Hours
Stress_Level_Nu...	Average	Avg_Stress_Level_Numeric
Physical_Activity...	Average	Avg_Physical_Activity_Hours
Lifestyle_Risk_In...	Average	Avg_Lifestyle_Risk_Index
Focus_Score	Average	Avg_Focus_Score
Productivity_Sco...	Average	Avg_Productivity_Score
Student_ID	Count	Count

Figure 7: Summarize Tool Configuration

8. Output Data Tool

   The Output Data tool was configured twice to export two downstream-ready files in CSV format. The first export, master_lifestyle_output.csv contains row-level student records for use in detail-level visualizations. The second export, summary_by_segment.csv contains the aggregated segment metrics structured specifically for group-level dashboard charts in Tableau or Power BI.

3.0 Challenges Faced

1. Heterogeneous naming conventions: Dataset 1 uses PascalCase (e.g., Study_Hours_Per_Day), Dataset 2 uses full lowercase snake_case (e.g., study_hours_per_day), and Dataset 3 uses partially consistent PascalCase with legacy labels such as CGPA. Without a strict schema alignment step using the Select tool, any downstream Join or Union would fail or produce incorrect output.

2. Dataset 2 does not contain a GPA field, resulting in null values appearing in the GPA_Band column for all records originating from that dataset following the Union merge. Excluding these records entirely would have significantly reduced the total record count and introduced analytical bias toward Dataset 1 exclusively. The challenge was therefore managed by allowing null

GPA_Band values to pass through the aggregation without exclusion, ensuring all available records contributed to the final output.